Yi Zheng

SENIOR AI RESEARCH SCIENTIST

Computer Vision • Multimodal AI • Biometrics • Medical AI • Graph Neural Networks

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Senior AI Research Scientist with a strong background in computer vision, multimodal learning, biometrics, and medical AI. Experienced in developing production-scale deep learning systems from research to deployment. Inventor of a U.S. patent-pending fingerprint matching framework and author of publications in top-tier venues including IEEE TMI and IJCV.

TECHNICAL SKILLS

AI & MACHINE LEARNING

Deep Learning (CNNs, Transformers, GNNs)
Computer Vision / Image Processing
Object Detection / Segmentation
OCR / Scene Text Recognition
Biometrics (Fingerprint, Face)
Multimodal Learning
Vision--Language Models
Large Language Models
Representation Learning
Medical Image Analysis
Graph Neural Networks
Statistical Graphical Models

PROGRAMMING & TOOLS

Python • C++ • CUDA
PyTorch • TensorFlow
OpenCV • MATLAB
Linux • Git

EDUCATION

Boston University — 2024
Ph.D. in Computer Science
GPA: 3.93/4.0

University of Southern California — 2016
M.S. in Electrical Engineering
GPA: 3.95/4.0

Shandong University — 2014
B.S. in Electrical Engineering
GPA: 3.90/4.0

AWARDS & HONORS

- Computer Science Research Excellence Award (REA), Boston University — 2022
- Masters Honors Fellowship, University of Southern California — 2015

PATENTS

- **U.S. Patent**
  Graph-Based Fingerprint Matching Network
  Thales DIS USA, Inc. — 2025
- **U.S. Patent**
  Image-Based Collimator Edge Detection
  GE Healthcare — 2017

WORK EXPERIENCE

Thales DIS USA, Inc. (Thales Group) — Los Angeles, CA

Senior AI Research Scientist — 2024 - Present

- Lead research and development of AI-powered biometric recognition systems for large-scale identity verification.
- Invented a graph-based fingerprint matching network (U.S. patent pending) leveraging graph neural networks, relational reasoning, and attention mechanisms to improve matching accuracy, robustness, and scalability in operational biometric systems.
- Developed a unified minutiae detection and pose estimation framework using multi-task learning and lightweight object detection architectures to jointly localize fingerprint minutiae and estimate ridge orientation, improving feature extraction accuracy and downstream matching performance.
- Optimized and deployed deep learning models for real-world production systems through inference acceleration, model compression, and large-scale system integration.

GE Healthcare — Beijing, China

Image Quality Engineer — Jun 2016 – Jul 2017

- Developed image processing and machine learning algorithms for X-ray and CT imaging systems.
- Designed the Image-Based Collimator Edge Detection (ICED) algorithm (patent granted) to improve automated analysis and system reliability.
- Implemented data-driven optimization and signal interpretation methods supporting downstream diagnostic workflows.

SELECTED PUBLICATIONS

J = Journal C = Conference W = Workshop

[J.1] **Y. Zheng**, et al., FourierMIL: Fourier filtering-based multiple instance learning for whole slide image analysis, *International Journal of Computer Vision*, 2026.

[J.2] L. Xu, Y. Kefella, **Y. Zheng**, et al., Attention-based deep learning for analysis of pathology images and gene expression data in lung squamous premalignant lesions, *Genome Medicine*, 2026.

[J.3] **Y. Zheng**, et al., Graph attention-based fusion of pathology images and gene expression for prediction of cancer survival, *IEEE Transactions on Medical Imaging*, 2024.

[J.4] R. Gindra, **Y. Zheng**, et al., Graph perceiver network for lung tumor and bronchial premalignant lesion stratification from histopathology, *The American Journal of Pathology*, 2024.

[J.5] **Y. Zheng**, et al., A deep learning-based graph-transformer for whole slide image classification, *IEEE Transactions on Medical Imaging*, 2022.

[J.6] **Y. Zheng**, et al., Deep-learning-driven quantification of interstitial fibrosis in digitized kidney biopsies, *The American Journal of Pathology*, 2021.

[C.1] **Y. Zheng**, et al., Semantic-Based Sentence Recognition in Images Using Bimodal Deep Learning, *IEEE International Conference on Image Processing (ICIP),* 2021.

[C.2] **Y. Zheng**, et al., LAL: Linguistically aware learning for scene text recognition, *ACM International Conference on Multimedia (ACM MM)*, 2020.

[W.1] Q. Wang, **Y. Zheng**, et al., A method for detecting text of arbitrary shapes in natural scenes that improves text spotting, *Computer Vision and Pattern Recognition Workshop (CVPR Workshop)*, 2020.

PROFESSIONAL SERVICE

Reviewer / Program Committee Member:

CVPR, ICCV, IEEE Transactions on Medical Imaging, Nature Communications, IEEE Access, ISBI, ISCAS, CMPB, Journal of Alzheimer's Disease, Micron, PETRA, IJIG, PBVS Workshop at CVPR (2021–2025).

TEACHING EXPERIENCE

- **Lead TA, CS 640: Artificial Intelligence**, Boston University (Fall 2017, Fall 2018)
- **Lead TA, CS 132: Linear Algebra**, Boston University (Spring 2018, Spring 2019)